1. Interpret a mathematical result back into the real-world context

Relevance: ★★★★★ (Highest)

- Students must take the calculated penguin population results
- And reinterpret them within the context of **real penguin colony development**
- Focusing on what these numbers actually **mean in reality**

👉 This is the **core purpose** of QS4: moving from "numbers" back to the "real world".

---

2. Identify whether a mathematical result or conclusion does, or does not, make sense given the context

Relevance: ★★★★★

- Students are required to evaluate questions such as:
  - Is the population growth unrealistically fast?
  - Is long-term exponential growth plausible?
- This involves **contextual judgement and common sense**, rather than checking algebraic correctness

👉 This reflects a classic **reasonableness-checking competence**.

---

3. Identify the limits of the model used to solve a problem

Relevance: ★★★★☆

- Students need to recognise that the model ignores:
  - Food supply limitations
  - Environmental carrying capacity
  - Disease, migration, and climate effects
- The model is therefore valid only **under restricted conditions and timeframes**

👉 This distinguishes **mathematical feasibility from ecological realism**.

---

4. Recognise the extent and limits of mathematical concepts and mathematical solutions

Relevance: ★★★★☆

- Exponential models are mathematically sound
- But they cannot be extrapolated indefinitely in real contexts
- This highlights the **boundaries of mathematical tools**

5. Understand the relationship between the context of the problem and representation of the mathematical solution

Relevance: ★★★★☆

- Students must understand:
  - How the exponential expression arises from the real-world assumptions
  - Whether the representation remains valid when real conditions change
- Context is used to critically assess the mathematical form

6. Use knowledge of how the real world impacts outcomes to make contextual judgements

Relevance: ★★★☆☆

- Requires basic ecological knowledge:
  - Animal populations cannot grow without limit
  - Environmental factors constrain survival
- Real-world understanding is used to **challenge or qualify the mathematical result**

---

7. Construct and communicate explanations and arguments in the context of the problem

Relevance: ★★★☆☆

- Students must clearly explain:
  - Why the result is reasonable or unreasonable
  - Which assumptions lead to unrealistic outcomes
- Emphasis is on **argumentation and explanation**, not calculation

---

8. Interpret mathematical outcomes in a variety of formats or compare representations

Relevance: ★★☆☆☆

- May involve comparing:
  - Numerical outputs with real-world implications
- However, multiple formal representations (e.g. graphs) are typically not required

---

9. Use mathematical tools or computer simulations to ascertain reasonableness

Relevance: ★☆☆☆☆ **(Lowest)**

- QS4 is primarily a **conceptual reflection task**
- No technological tools or simulations are required
- The focus is on reasoning and judgement

① Interpret a mathematical result back into the real world context

（将数学结果解释回现实情境）

相关性：★★★★★（最高）

- 学生需要将得到的企鹅数量变化结果
- 重新放回到**真实企鹅种群发展**的背景中理解
- 关注"这些数字在现实中意味着什么"

👉 这是 QS4 的**核心目标**：从"数"回到"世界"。

---

② Identify whether a mathematical result or conclusion does, or does not, make sense given the context

（判断结果在情境中是否合理）

相关性：★★★★★

- 要评估：
  - 种群是否增长过快？
  - 长期指数增长是否现实？
- 强调**常识 + 情境判断**，而非公式正确性

👉 这是典型的 **"结果合理性检验"能力。**

---

③ Identify the limits of the model used to solve a problem

（识别模型的局限性）

相关性：★★★★☆

- 学生需要意识到模型忽略了：
  - 食物资源限制
  - 环境承载量
  - 疾病、迁徙、气候变化等因素
- 认识到模型只在**一定条件和时间范围内有效**

👉 明确区分 **"数学上可行" 与 "现实中可持续"。**

④ Recognise the extent and limits of mathematical concepts and solutions

（认识数学解的适用范围）

相关性：★★★★☆

- 指数模型在数学上是成立的
- 但并不意味着可以无限外推
- 强调数学工具的**能力边界**

⑤ Understand the relationship between the context of the problem and representation of the mathematical solution

（理解情境与数学表示之间的关系）

相关性：★★★★☆

- 学生需要理解：
  - 指数表达式如何来源于现实假设
  - 当现实条件改变时，数学表示是否仍然适用
- 通过情境反推公式的合理性

---

⑥ Use knowledge of how the real world impacts outcomes to make contextual judgments

（利用现实知识对结果作判断）

相关性：★★★☆☆

- 需要基本生态常识：
  - 动物种群不可能无限增长
  - 生存环境会制约数量
- 用现实知识**修正或质疑数学结果**

---

⑦ Construct and communicate explanations and arguments in context

（在情境中构建解释与论证）

相关性：★★★☆☆

- 要用语言清楚说明：
  - 为什么结果合理 / 不合理
  - 哪些假设导致偏差
- 强调**解释能力而非计算能力**

⑧ Interpret mathematical outcomes in a variety of formats or compare representations

（多表征解释或比较）

相关性：★★☆☆☆

- 可能涉及：
  - 数字结果 vs. 现实意义
- 但通常不要求图像或多种形式的比较

---

⑨ Use mathematical tools or computer simulations to test reasonableness

（使用技术检验合理性）

相关性：★☆☆☆☆**（最低）**

- QS4 通常是**概念性反思题**
- 不要求计算工具或模拟
- 更偏向思维与判断